



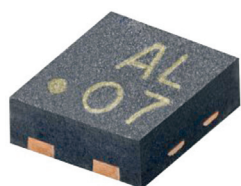
applications. The ML7670 receiver IC delivers a maximum power transfer efficiency of 45 % within the 250mW output range while maintaining a small package size.

ROHM Semiconductor

<https://www.rohm.com>

## Sensors for healthcare, wearables, and IoT

Murata Manufacturing has begun mass production of its MRMS166R and MRMS168R anisotropic magnetoresistance (AMR) sensors for healthcare, wearable, and IoT applications. The MRMS166R is the first AMR sensor to



reach 20nA average current consumption while operating at 1.2V supply, extending bat-

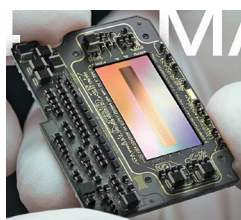
tery life in coin-cell-powered devices. These solid-state magnetic switches detect the presence or absence of a magnetic field and output a logic signal, enabling automatic transitions between active and sleep modes without mechanical contact, thereby supporting compact, sealed designs. This type of automatic power-state switching is used in battery-powered products to reduce standby power consumption and extend operating life.

Murata Manufacturing

<https://www.murata.com>

## Colour lidar sensor for object detection

Ouster has introduced Rev8, a native colour lidar sensor built on L4 ouster silicon for perception systems. It combines colour and 3D spatial data



using physical principles and Fujifilm colour science, enabling high-resolution output, low latency, and spatial-

temporal alignment. The system is based on combining geometry and colour for scene interpretation, allowing a single lidar sensor to identify road signs, brake lights, and produce colourised 3D maps. Rev8 is designed

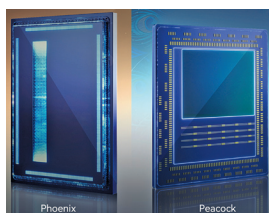
to generate large volumes of colour 3D data for physical AI systems, support perception pipeline simplification, and assist model training and deployment. It is intended for use in autonomy, infrastructure, and industrial environments, alongside the OS1 Max, which is used for long-range object detection and scene capture, supporting object classification and sensor fusion in system design.

Ouster

<https://ouster.com>

## SoC for high-density 3D perception

Peacock is RoboSense's fully solid-state SPAD-SoC designed to deliver high-density 3D perception for robotics and cross-industry applications. Built around a 640 × 480 sensor array, it transforms lidar outputs from sparse



point clouds into dense, VGA-level 3D imaging, improving environmental

understanding and spatial resolution. The chipset offers a 180° × 135° field of view, millimetre-level precision, and the ability to detect objects as close as 5cm, making it suitable for dynamic and close-range scenarios. By integrating a sensing matrix into a single chip, Peacock enables compact and scalable deployment across use cases, including robotics, industrial safety systems, low-speed autonomous vehicles, and automotive blind-spot detection.

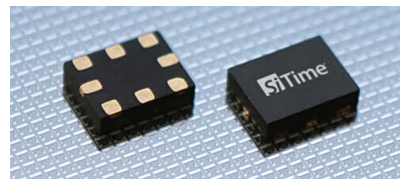
RoboSense

<https://www.robosense.ai>

## Precision timing solution for AI data centres

SiTime Corporation has introduced the Elite 2 Super-TCXO, a precision timing solution aimed at improving GPU utilisation and computational efficiency in AI data centres by enabling highly accurate time synchronisation; it supports sub-nanosecond synchronisation across AI compute clusters to reduce timing drift that can limit performance,

while delivering up to 100x better timing accuracy, 25x improved temperature stability, 8x lower Allan deviation, and 4x enhanced frequency



stability, all in a footprint reduced by up to 2x. It also adds digital frequency tuning, significantly lower jitter than quartz-based alternatives, and improved resistance to shock, vibration, and other mechanical stresses at the board level.

SiTime Corporation

<https://www.sitime.com>

## AI-based codec to reduce power consumption and memory needs

Ambiq Micro has announced compressionKIT, a next-generation AI-based codec currently in beta, designed to



significantly reduce power consumption and memory requirements when processing continuous sensor data in wearable and edge devices. As always-on systems such

as medical wearables, smart home devices, and industrial sensors generate constant data streams, the resulting storage and transmission demands have become a major constraint on battery life, memory capacity, and overall system efficiency. CompressionKIT addresses this by compressing sensor data at the source while preserving the key features needed for AI processing and analytics, enabling more efficient operation with reduced resource use. It delivers up to 20x higher efficiency in data handling and up to 16x lower memory usage compared to raw sensor streams.

Ambiq Micro

<https://www.ambiq.com>